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Population genetic analyses of seven Chinese indigenous chicken breeds in a context of global breeds

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Summary

Jiangxi province in China is rich in indigenous chicken breeds, which have diverse phenotypes and represent a valuable genetic resource for further genetic improvement of modern breeds. Here, we conducted a series of analyses to reveal genetic diversity, phylogenetic relationships and population structure of seven chicken breeds in Jiangxi province in the context of nine non-local chicken breeds, using 600K SNP data. We show that Jiangxi indigenous breeds have more abundant nucleotide diversity than do European local and commercial breeds. Among Jiangxi breeds, Dongxiang Blue-eggshell (DX) and Chongren Partride (CR) display remarkably reduced genetic diversity, as the two breeds exhibit increased inbreeding coefficients, runs of homozygosity, extent of linkage disequilibrium and reduced expected heterozygosity. DX, CR and Taihe Silkie (TH) represent three ancestral lineages of the Jiangxi chicken and display genetic differentiation from the other four Jiangxi breeds, which show a signature of admixture with European commercial breeds. These findings provide insight for the establishment of an efficient conservation program for Jiangxi chicken breeds. Considering the current status of genetic diversity and ancestral representativeness, particular attention should be paid to DX, CR and TH chickens.

Keywords China, Chinese indigenous chickens, conservation, genetic diversity, local chickens, phylogenetic relationship, population structure

China has a wealth of aboriginal chicken breeds that possess diverse traits adapted for local environments and low input-output production systems (China National Commission of Animal Genetic Resources 2011). Nine chicken breeds were originally distributed in Jiangxi province of China: Anyi Gray (AY), Chongren Partridge (CR), Dongxiang Blue-eggshell (DX), Guangfeng Baier (GF), Ningdu Yellow (ND), Taihe Silkie (TH), Xiushui Black-bone (XS), Wanzai Yellow and Yugan Black. These breeds provide a valuable genetic resource for scientific research and further genetic improvement in the chicken industry. For instance, TH has one of few skin pigmentation variants in the chicken with the unique phenotype of fibromelanosis (dermal hyperpigmentation) and silky-feather, which has been a subject of cultural importance and scientific interest for centuries (Dunn & Jull 1927; Dorshorst *et al.* 2011).

Currently, effective population sizes of the Jiangxi local chicken are decreasing due to lack of planning and extensive hybridization between modern and local breeds. To address this problem, the Chinese central government has launched a national conservation program for 28 Chinese chicken breeds including two Jiangxi breeds (GF and TH). To establish an efficient conservation strategy for Jiangxi local chickens, we herein investigated genetic diversity, population structure and phylogenetic relationships of seven Jiangxi breeds using the Affymetrix Axiom 600K Chicken Genotyping Array (Kranis *et al.* 2013).

We isolated genomic DNA from blood samples of 95 chickens from seven Jiangxi breeds (Table S1, Fig. 1a) using a standard phenol/chloroform method. Genomic DNA was genotyped for 580 961 SNPs using the Affymetrix Axiom 600K Chicken Genotyping Array. We downloaded the 600K SNP data of 140 chickens from seven local Eurasian breeds and two commercial breeds (Table S1) that were reported in previous studies (Ma *et al.* 2015; Pengelly *et al.* 2016). Acquired SNP datasets were formatted for analysis

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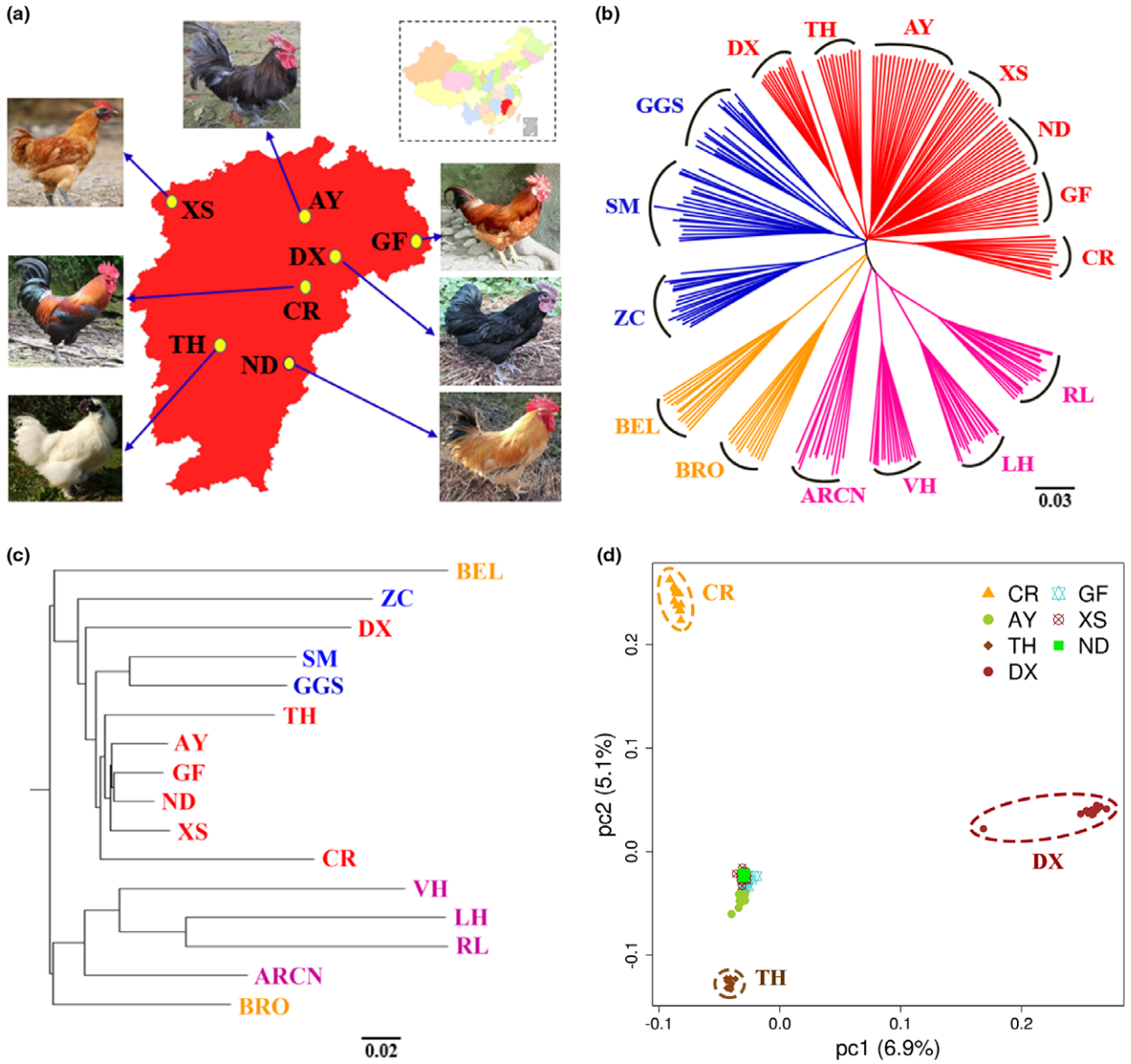


Figure 1 Sample location and genetic relatedness. (a) Geographic locations of Jiangxi chicken breeds. (b) Neighbor-joining tree based on an identical-by-state matrix. (c) Neighbor-joining tree based on genetic differentiation (F_{ST}) values among breeds. (d) Principal component plot of Jiangxi chicken breeds. Abbreviations of all breeds are given in Table S1.

using PLINK v 1.90 (Purcell *et al.* 2007). Individuals with a call rate of greater than 0.95 (--mind 0.05) and SNPs with a call rate of greater than 0.99 (--geno 0.01), minor allele frequency of greater than 0.01 (--maf 0.01) and Hardy–Weinberg equilibrium (HWE) test $P < 10^{-6}$ (--hwe 0.000001) were retained. All unmapped SNPs and those on sex chromosomes were excluded. A final set of 221 463 SNPs from 235 chickens was used for the subsequent analyses.

We calculated expected heterozygosity (H_E) using the PLINK '--hardy' command. The inbreeding coefficient (F) was calculated using the PLINK --het' command. Runs of homozygosity (ROHs) were also calculated using PLINK v 1.90 (Purcell *et al.* 2007) with a sliding window approach: 50-SNP sliding windows that allowed one heterozygote and five missing calls per window. To exclude short and very common ROHs, the minimum length for an ROH segment was set at 500 kb. Linkage disequilibrium (LD, r^2) between pairs of SNPs was estimated using the PLINK command '--r^2-ld-window-kb -ld-window-r^2 0'. Individual pairwise genetic distance was calculated using PLINK v 1.90 (Purcell *et al.* 2007). Weir & Cockerham's (1984) average F_{ST} was explored to measure the genetic distance between each pair of breeds. Neighbor-joining (NJ) phylogenetic trees were then constructed using the NEIGHBOR program implemented in PHYLIP v3.695 (Felsenstein 1989), and the

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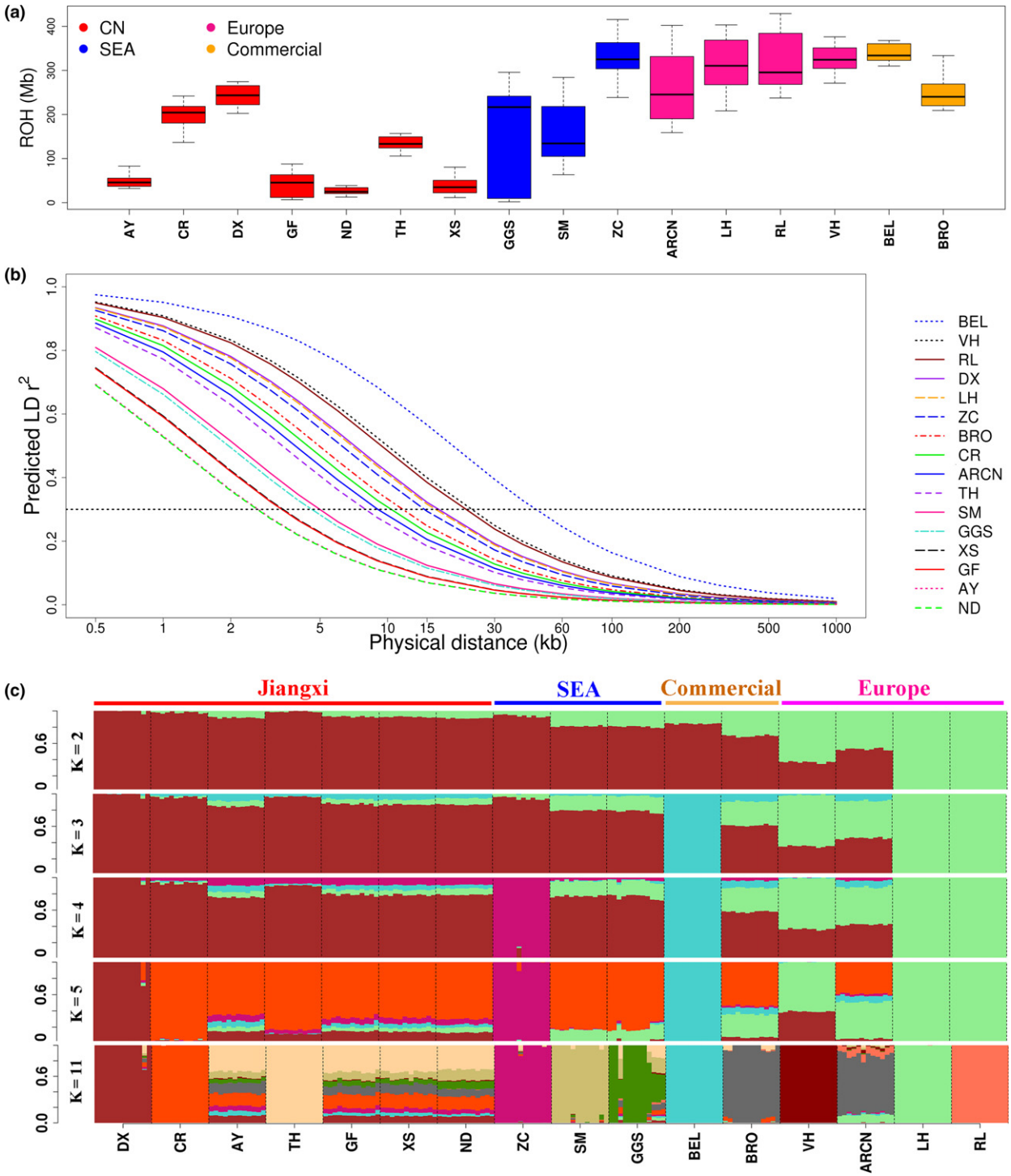


Figure 2 Genetic diversity and population structure. (a) Runs of homozygosity, each point denoting the ROH value of each individual. (b) Extents of LD (predicted r^2) plotted as a function of inter-SNP distance for all breeds. Inter-SNP distances (kb) were binned into the following classes: 0–1, 1–2, 2–4, 4–6, 6–9, 9–15, 15–30, 30–40, 40–60, 60–80, 80–100, 100–200, 200–300, 300–500, 500–800, 800–1000 kb. Observed pair-wise LD (r^2) was averaged for each inter-SNP distance class. The black dotted line indicates the threshold of 0.3. (c) Ancestry compositions of the tested breeds. Twelve individuals from each breed were used for this analysis. Each color represents one ancestral cluster. Abbreviations for all breeds are given in Table S1.

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resultant trees were displayed via FIGTREE v1.4.2 (Rambaut 2014). Principal component analysis (PCA) was performed using GCTA (Yang *et al.* 2011). To avoid potential artifacts due to LD, we used 159 169 SNPs for which LD was less than 0.5. ADMIXTURE 1.23 (Alexander *et al.* 2009) was used to calculate maximum likelihood estimates of the most probable number of ancestral populations (K) based on data on the 159 169 SNPs.

Of the seven Jiangxi breeds, four (AY, GF, XS and ND) had greater H_E values ($H_E = 0.33–0.34$) than did European ($H_E = 0.22–0.30$) and commercial ($H_E = 0.25–0.31$) breeds. The other three Jiangxi breeds (CR, TH and DX) had H_E values of 0.26–0.29, which are comparable to those of the nine non-local breeds (Table S1). In general, Jiangxi breeds have smaller ROHs and F-values than do European and commercial breeds. Among the Jiangxi breeds, DX had the largest ROH (232 Mb) and F-value (0.237), followed by CR with a ROH of 204 Mb and an F-value of 0.223. AY, GF, XS and ND showed much lower ROH and F-values than did the other breeds (Table S1, Fig. 2a). The extent of LD was generally longer in European and commercial breeds than in Jiangxi breeds. Among the Jiangxi breeds, DX had the longest extent of LD ($r^2_{0.3} = 16.64$ kb), followed by CR ($r^2_{0.3} = 10.27$ kb). AY ($r^2_{0.3} = 2.63$ kb), GF ($r^2_{0.3} = 3.37$ kb), XS ($r^2_{0.3} = 3.41$ kb) and ND ($r^2_{0.3} = 2.61$ kb) displayed much lower LD levels compared with the non-local breeds (Table S1, Fig. 2b). These findings show that Jiangxi indigenous chickens are richer in genetic variability than are European and commercial breeds, which is in agreement with a previous study (Qu *et al.* 2006). However, CR and DX chickens showed reduced nucleotide variability. This could be explained by the fact that the two breeds have recently experienced intensive selection for meat (CR) and egg (DX) production and could have undergone inbreeding to a certain degree.

In the NJ tree based on genome-wide allele sharing (Fig. 1b), all individuals from the same breed clustered together. Jiangxi and European/commercial chickens defined two separate clades. Three Southeast Asian breeds (*Gallus gallus spadiceus*, Shamo and Zwerg-Cochin) showed a closer phylogenetic relationship with Jiangxi breeds. The clustering pattern in this tree is consistent with that in the F_{ST} estimate-derived dendrogram, except that the two commercial breeds were separated into Asian and European clades (Fig. 1c).

The PCA results illustrate that European, Southeast Asian, Jiangxi and commercial breeds reflect four groupings. PC1 clearly separated European breeds from Asian and commercial breeds, and PC2 displayed the genetic differentiation between commercial and Asian breeds (Fig. S1a). PC3 and PC4 identified Southeast Asian and Jiangxi breeds in different clusters (Fig. S1b). When we focused on Jiangxi breeds, PC1 and PC2 separated DX, CR and TH from the other four breeds (Fig. 1d). PC3 and PC4 identified AY as being in a different region, whereas GF, ND and XS did not form separate clusters (Fig. S1c), which is in agreement with the NJ and F_{ST} clustering results.

The results of the ADMIXTURE analysis for the16 tested breeds with K values from 2 to 5 are shown in Fig. 2c, and the lowest cross-validation error was found at $K = 11$ (Fig. S2). When $K = 2$, Jiangxi and European breeds appeared as two differentiated clusters, whereas variable fractions of European ancestry were evidenced in four of the Jiangxi local breeds (AY, GF, ND and XS) and the three Southeast Asian breeds. When $K = 3–5$, two ancestral lineages were observed in Jiangxi breeds. DX represented a distinct lineage, whereas CR displayed another ancestral lineage from which the other five breeds (AY, GF, ND, TH and XS) mainly derived. When $K = 11$, TH defined a separate group whereas AY, GF, ND and XS were displayed as having similar ancestry structure. The two European breeds (Araucana and Vorwerkhühner) had a mixture of European and Asian lineages, which is in agreement with their breeding history and previous reports (Gongora *et al.* 2008; Wilkinson *et al.* 2012). The two commercial breeds (Brown egg layer and Broiler) had the same genetic makeup as did Chinese breeds ($K = 2$), as previously reported (Lyimo *et al.* 2015).

We show that AY, GF, XS and ND have a closer phylogenetic relationship with each other than with any other breeds, as they always clustered together in both the NJ trees and the PCA plots and exhibited similar ancestral composition patterns. This indicates that the four breeds likely derived from a common ancestral population. We note that the four breeds had a signature of admixture with European and commercial breeds in the ADMIXTURE analysis (Fig. 2c), which may account for more abundant genetic diversity in these breeds. DX, CR and TH exhibit a clear divergence from the other four Jiangxi breeds and represent old ancestral lineages of Jiangxi local chickens. From a conservation point of view, DX, CR and TH should top the priority list. Considering the current status of genetic diversity, particular attention should be paid to DX and CR, as they show reduced nucleotide variability and increased inbreeding coefficients. An elaborate conservation program must be established for the two breeds to avoid further inbreeding and to maintain their nucleotide diversity.

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Supporting information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Figure S1 Principal component plots.

Figure S2 Cross-validation errors of the ADMIXTURE analysis at different *K* values.

Table S1 Genetic diversity of 16 tested breeds in this study.

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